Using a saddle point approximation, the action can be expanded near it's minima to a quadratic polynomial, in order to solve the integrals appearing in the matrix elements of $\tilde{Q}$. This results in two equations for b, c which can be solved numerically.

A.4 INTEGRAL CALCULATIONS FOR THE TEACHER-STUDENT MODEL

Here we compute the integral $\left\langle (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n+1} \phi(\boldsymbol{w} \cdot \boldsymbol{x}) \right\rangle_{\boldsymbol{x}}$ where $\boldsymbol{x} \sim \mathcal{N}(0, I_d)$. This integral appears both in the computation of the action and in the calculation of the matrix elements of $\tilde{Q}$. Denote-

$$I_n \equiv \frac{1}{Z} \int d\boldsymbol{x} e^{-\frac{1}{2}\boldsymbol{x}^2} \phi(\boldsymbol{w} \cdot \boldsymbol{x}) (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n+1}. \tag{13}$$

Where $Z = (2\pi)^{d/2}$ is a normalizing factor. This integral is given by

$$\begin{aligned} I_n &\equiv \frac{1}{Z} \int d\boldsymbol{x} e^{-\frac{1}{2}\boldsymbol{x}^2} \phi(\boldsymbol{w} \cdot \boldsymbol{x}) (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n+1} \\ &= \frac{2}{\sqrt{\pi}} \frac{\boldsymbol{w} \cdot \boldsymbol{w}^*}{\sqrt{\det(\mathbf{1} + 2\boldsymbol{w}\boldsymbol{w}^T)}} \left\langle (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n} \right\rangle_{\mathcal{N}\left(0, (\mathbf{1}+2\boldsymbol{w}\boldsymbol{w}^T)^{-1}\right)} + 2n |\boldsymbol{w}^*|^2 I_{n-1} \\ &= \frac{2}{\sqrt{\pi}} \frac{\boldsymbol{w} \cdot \boldsymbol{w}^*}{\sqrt{1 + 2|\boldsymbol{w}|^2}} \left\langle (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n} \right\rangle_{\mathcal{N}\left(0, (\mathbf{1}+2\boldsymbol{w}\boldsymbol{w}^T)^{-1}\right)} + 2n |\boldsymbol{w}^*|^2 I_{n-1} \end{aligned} \tag{14}$$

By Sherman Morrison's formula,

$$\left(\mathbf{1} + 2\boldsymbol{w}\boldsymbol{w}^T\right)^{-1} = \mathbf{1} - \frac{2\boldsymbol{w}\boldsymbol{w}^T}{1 + 2|\boldsymbol{w}|^2}. \tag{15}$$

Therefore, a recursive expression for I_n can be obtained in terms of $\boldsymbol{w}$ and I_{n-1}

$$\begin{aligned} I_n &= \frac{1}{Z} \int d\boldsymbol{x} e^{-\frac{1}{2}\boldsymbol{x}^2} \phi(\boldsymbol{w} \cdot \boldsymbol{x}) (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n+1} \\ &= \frac{2}{\sqrt{\pi}} \frac{\boldsymbol{w} \cdot \boldsymbol{w}^*}{\sqrt{1 + 2|\boldsymbol{w}|^2}} \left\langle (\boldsymbol{w}^* \cdot \boldsymbol{x})^{2n} \right\rangle_{\mathcal{N}\left(0, (\mathbf{1}+2\boldsymbol{w}\boldsymbol{w}^T)^{-1}\right)} + 2n |\boldsymbol{w}^*|^2 I_{n-1}. \end{aligned} \tag{16}$$

The cases $n = 0, 1$ are the ones used in this work. For $n = 0$ the integral is simply-

$$I_0 = \frac{2}{\sqrt{\pi}} \frac{\boldsymbol{w} \cdot \boldsymbol{w}^*}{\sqrt{1 + 2|\boldsymbol{w}|^2}}. \tag{17}$$

For $n = 1$ the calculation is slightly more complicated. We have

$$\begin{aligned} \left\langle (\boldsymbol{w}^* \cdot \boldsymbol{x})^2 \right\rangle &= \sum_{ij} w_i' w_j' \langle x_i x_j \rangle = \sum_{ij} w_i' w_j' \left[\left(\mathbf{1} + 2\boldsymbol{w}\boldsymbol{w}^T\right)^{-1}\right]_{ij} \\ &= \sum_{ij} w_i' w_j' \left[\delta_{ij} - \frac{2 w_i w_j}{1 + 2|\boldsymbol{w}|^2}\right] \\ &= |\boldsymbol{w}^*|^2 - \frac{2(\boldsymbol{w} \cdot \boldsymbol{w}^*)^2}{1 + 2|\boldsymbol{w}|^2}. \end{aligned} \tag{18}$$

Which yield,

$$I_1 = \frac{2}{\sqrt{\pi}} \frac{(\boldsymbol{w} \cdot \boldsymbol{w}^*)}{\sqrt{\left(1 + 2|\boldsymbol{w}|^2\right)}} \left(3|\boldsymbol{w}^*|^2 - \frac{2(\boldsymbol{w} \cdot \boldsymbol{w}^*)^2}{1 + 2|\boldsymbol{w}|^2}\right) \tag{19}$$

B MODULAR ALGEBRA

B.1 MODULAR ALGEBRA KERNEL AND SYMMETRIES

We are first concerned with the spectral properties of $Q_{nm,n'm'}$, the latter being the average of $\tilde{Q}_{nm,n'm'}$ with respect $S[\boldsymbol{w}]$. To this end note some useful model symmetries that occur when n_{train} spans the entire training set. By symmetries here we mean that the training algorithm, neural network, and the data-set are invariant remain invariant under the following transformations:

I. Taking $[n, m] \to [(n+q) \bmod P, (m+q') \bmod P]$, and $f_p \to f_{(p+q+q') \bmod P}$ with $q, q' \in \mathbb{Z}_P$

II. Taking $[n, m] \to [qn \bmod P, qm \bmod P]$ and $f_p \to f_{qp \bmod P}$ for $q \in \mathbb{Z}_P$ but different than zero.

Notably, Q does not contain any reference to a specific label (p). Hence, both these symmetries are symmetries of Q (both before and after any potential phase transition). Specifically, let us denote by T_1 (T_2) the orthogonal transformation taking $[n, m] \to [n+1, m] \bmod P$ ($[n, m] \to [n, m+1] \bmod P$) and by C_q the transformation taking $[n, m] \to [qn, qm] \bmod P$. Then the following commutation relations holds,

$$
\begin{aligned}
[Q, T_1] = [Q, T_2] = [Q, C_q] &= 0 \\
[T_1, T_2] &= 0
\end{aligned} \tag{20}
$$

This implies that Q, T_1 and T_2 can be diagonalized on the same basis. The Fourier basis ($\phi_{k,k'}$), discussed in the main text, clearly diagonalizes T_1 and T_2 simultaneously. Furthermore, as no two $\phi_{k,k'}$ share the same eigenvalues of T_1 and T_2, each $\phi_{k,k'}$ must be an eigenfunction of Q.

Next, we notice that since $[Q, C_q] = 0$, $C_q \phi_{k,k'}$ must also be an eigenvector of Q with the same eigenvalue. However, since $[T_{1,2}, C_q] \neq 0$ C_q shifts us between, $\phi_{k,k'}$ hence implying degenerates. Based on the definition of the Fourier modes, we find $C_q \phi_{k,k'} = \phi_{qk,qk'}$ hence all pairs of momentum (k, k',g, g') such that $k = qg, k' = qg'$ must have degenerate eigenvalues. Notably for prime P for any $k, g \neq 0$ there exists a q such that $qk = g$. We furthermore note that an additional "flip" symmetry exists, implying that k, k' are associated with the same eigenvalue as k', k.

The above results for Q hold exactly regardless of network width, noise, or other hyperparameters. Next, we focus on Q at the NNGP limit ($N \to \infty$), which according to our saddle-point treatment, is also the kernel at any point before the phase transition (at large P). Here we find an enhanced symmetry which explains its poor performance. Indeed, before the phase transition, The kernel Q is given

$$
\begin{aligned}
Q_{nm,n'm'} &= \sigma_a^2 \left[K_{nm,nm} K_{n'm',n'm'} + 2K^2_{nm,n'm'} \right] \\
K_{nm,n'm'} &= \langle (W_n + W_{m+P})(W_{n'} + W_{m'+P}) \rangle_{\boldsymbol{w} \sim N[0, I_{2P \times 2P}]} \\
&= \delta_{n,n'} + \delta_{m+P,m'+P} + \delta_{n,m'+P} + \delta_{m+P,n'} \\
&= \delta_{n,n'} + \delta_{m,m'}
\end{aligned} \tag{21}
$$

where we recall that $n, m \in [0...P-1]$. Given that, one can act with symmetry II separately on each index. This extra symmetry implies, via the above arguments, that all k, k', and q, q' with $k = ck'$ and $q = c'q'$ are equivalent. For prime, P this results in only 3 distinct eigenvalues - the one associated with $k, k' = 0, 0$, the one associated with all $0, k \neq 0$ and $k \neq 0, 0$ modes and the one (λ) associated with $k \neq 0, k' \neq 0$. Notably, the target is supported only on this last $(P-1)^2$-dimensional degenerate subspace. The equal prior on $(P-1)^2$ eigenfunctions then implies $O(P^2)$ examples are needed for the GP to perform well on the task here, namely the entire multiplication table.

B.2 FOURIER TRANSFORM OF THE ACTION

Here we compute the action in Fourier space without the weight decay term which appears in the main text. To construct the Fourier transformed action, we start from the fluctuating kernel term ($\tilde{Q}$)

given by

$$\sigma_a^{-2}\tilde{Q} = \sum_{n,n'=0}^{P}\sum_{m,m'=P}^{2P-1} \left(w_m + w_n\right)^2 \left(w_{m'} + w_{n'}\right)^2 |n,m\rangle \langle n',m'| \tag{22}$$

where we have used physics-style bra-ket notation to denote the basis on which the operator $\tilde{Q}$ is presented. For instance, $\tilde{Q}_{n_0 m_0, n_0' m_0'}$ is given by acting with $\langle n_0, m_0|$ from the left and $|n_0', m_0'\rangle$ from the right and using $\langle n_0, m_0|n_1, m_1\rangle = \delta_{n_0,n_1}\delta_{m_0,m_1}$.

Based on the symmetries of the problem we consider the following coordinate transformation

$$w_m \mapsto \sum_{k=0}^{P-1} w_k e^{-\frac{2\pi i}{P}kn}, \quad w_n \mapsto \sum_{k=0}^{P-1} v_k e^{-\frac{2\pi i}{P}km} \tag{23}$$

Substituting this in the equation for $\tilde{Q}$:

$$\begin{aligned}\tilde{Q} = \sum_{kk'qq'} &\underbrace{\sum_{m,n}\left(w_k e^{-\frac{2\pi i}{P}kn} + v_k e^{-\frac{2\pi i}{P}km}\right)\left(w_{k'} e^{-\frac{2\pi i}{P}k'n} + v_{k'} e^{-\frac{2\pi i}{P}k'm}\right)|n,m\rangle}_{J} \\ &\cdot \underbrace{\sum_{m',n'}\left(w_q e^{\frac{2\pi i}{P}qn'} + v_q e^{\frac{2\pi i}{P}qm'}\right)\left(w_{q'} e^{\frac{2\pi i}{P}q'n'} + v_{q'} e^{\frac{2\pi i}{P}q'm'}\right)\langle n',m'|}_{J'}\end{aligned} \tag{24}$$

Where we can simplify further

$$\begin{aligned}J &= \sum_{mn}\left(w_k w_{k'} e^{-\frac{2\pi i}{P}\left(k+k'\right)n} + v_k v_{k'} e^{-\frac{2\pi i}{P}\left(k+k'\right)m} + w_{k'} v_k e^{-\frac{2\pi i}{P}\left(k'n+km\right)} + w_k v_{k'} e^{-\frac{2\pi i}{P}\left(kn+k'm\right)}\right)|n,m\rangle \\ &\equiv P\left(w_k w_{k'}\,|k+k',0\rangle + v_k v_{k'}\,|0,k+k'\rangle + w_k v_{k'}\,|k,k'\rangle + w_{k'} v_k\,|k',k\rangle\right),\end{aligned} \tag{25}$$

(which also serves as the definition of the Fourier transformed bra-ket notation) and similarly for the conjugated term:

$$J' = P\left(\langle -q,-q'|\, w_q v_{q'} + \langle -q',-q|\, w_{q'} v_q + \langle -\left(q+q'\right),0|\, w_q w_{q'} + \langle 0,-(q+q')|\, v_q v_{q'}\right) \tag{26}$$

We next turn to the quadratic term

$$\begin{aligned}\sum_{n=0}^{P-1} w_n^2 &= \sum_{n=0}^{P-1}\sum_{k,k'=0}^{P-1} w_k w_{k'} e^{\frac{2\pi i}{P}\left(k+k'\right)n} \\ &= P\sum_{k=0}^{P-1} w_k w_{-k},\end{aligned} \tag{27}$$

using $\sum_{n=0}^{P-1} e^{\frac{2\pi i}{P}\left(k+k'\right)n} = P\delta_{k,-k}$. The quadratic term in the action (main text equation 3.2) $|\boldsymbol{w}|^2$ is given by

$$|\boldsymbol{w}|^2 = \sum_{n=0}^{P-1} w_n^2 + \sum_{m=P}^{2P-1} w_m^2 = P\left(\sum_{k=0}^{P-1} w_k w_{-k} + \sum_{k=0}^{P-1} v_k v_{-k}\right) \tag{28}$$

The target can be also expressed in terms of the Fourier basis, noting that the mode zero is zero for P prime

$$y_{nm}^p = P^{-1}\sum_{k=1}^{P-1} e^{-i2\pi kp/P} e^{i2\pi k(n+m)/P} = \langle n,m|\sum_{q=1}^{P-1} e^{-2\pi iqp/P}\,|q,q\rangle \tag{29}$$

The remaining term in the action (main text equation 23) following the anzats $\sigma^2 t = ay$, and considering the whole dataset. the involves the kernel and is given by

$$\sum_{p,mn,m'n'} y^p_{nm} \overline{y}^p_{n'm'} \tilde{Q}_{nmn'm'} \tag{30}$$

$$= \sum_{p,mn,m'n'} \langle n, m| \sum_{q=1}^{P-1} e^{-2\pi iqp/P} |q, q\rangle \langle n', m'| \sum_{q'} e^{2\pi iq'p/P} |q', q'\rangle \tilde{Q}_{nmn'm'}$$

$$= \sum_{mn,m'n'} \sum_{q,q'=1}^{P-1} \langle n, m \mid q, q\rangle \langle n', m' \mid q', q'\rangle \tilde{Q}_{nn'mm'} \sum_{p} e^{-2\pi i\left(q'-q\right)p/P}$$

$$= P \sum_{mn,m'n'} \sum_{q,q'} \langle n, m \mid q, q\rangle \langle n', m' \mid q', q'\rangle \tilde{Q}_{nn'mm'} \delta_{q,q'}$$

$$= P \sum_{q} \langle q, q| \left[\sum_{mn,m'n'} \tilde{Q}_{nn'mm'} |n, m\rangle \langle n', m'| \right] |q, q\rangle$$

$$= P \sum_{q=1}^{P-1} \langle q, q| \tilde{Q} |q, q\rangle$$

Note that the chosen basis is orthonormal, so that $\langle t \mid k\rangle = \delta_{k,t}$ and thus all terms with $|0\rangle$ vanish and, and we are left with-

$$P \sum_{t=1}^{P-1} \langle t, t| \tilde{Q} |t, t\rangle \tag{31}$$

$$= P^3 \sum_{kk'qq'} \sum_{t=1}^{P-1} \langle t, t| \left(w_k v_{k'} |k, k'\rangle + w_{k'} v_k |k', k\rangle\right) \left(w_q v_{q'} \langle -q, -q'| + w_{q'} v_q \langle -q', -q|\right) |t, t\rangle$$

$$= P^3 \sum_{kk'qq'} \sum_{t=1}^{P-1} \left(w_k v_{k'} \delta_{k,t} \delta_{k',t} + w_{k'} v_k \delta_{k,t} \delta_{k',t}\right) \left(w_q v_{q'} \delta_{-q,t} \delta_{-q',t} + w_{q'} v_q \delta_{-q,t} \delta_{-q',t}\right)$$

$$= 4P^3 \sum_{t=1}^{P-1} w_t v_t w_{-t} v_{-t}.$$

The action in Fourier space is then given by

$$\mathcal{S}[\hat{\boldsymbol{w}}] = P \left[\left(\frac{1}{2} \sum_{k=0}^{P-1} w_k w_{-k} + \frac{1}{2} \sum_{k=0}^{P-1} v_k v_{-k} \right) - \frac{2\sigma_a^2 a^2 P^2}{N\sigma^4} \sum_{k=1}^{P-1} w_k w_{-k} v_k v_{-k} \right]. \tag{32}$$

Note that since the weights are all real, the Fourier transforms obey $\overline{w_k} = w_{-k}$, we can therefore rewrite $\mathcal{S}$ in terms of the parameters $|w_k|, |v_k|$

$$\mathcal{S}\left[\hat{\boldsymbol{w}}\right] = P \left[\frac{1}{2} \sum_{k=0}^{P-1} \left(|w_k|^2 + |v_k|^2 \right) - \frac{2\sigma_a^2 a^2 P^2}{N\sigma^4} \sum_{k=1}^{P-1} |w_k|^2 |v_k|^2 \right]. \tag{33}$$

So that the action can be separated into independent actions $\mathcal{S}_k$ that each depend on different values of k

$$\mathcal{S}[\hat{\boldsymbol{w}}] = \sum_{k\neq 0}^{P-1} P \left[\frac{1}{2} \left(|w_k|^2 + |v_k|^2 \right) - \frac{2\sigma_a^2 a^2 P^2}{N\sigma^4} |w_k|^2 |v_k|^2 \right] + \frac{1}{2} \left(|w_0|^2 + |v_0|^2 \right)$$

$$\equiv \mathcal{S}_0(|w_0|, |v_0|) + \sum_{k\neq 0}^{P-1} \mathcal{S}_k(|w_k|, |v_k|) \tag{34}$$

B.3 Solving the equation for a

As done in the teacher-student model, obtaining a self-consistent equation for the discrepancy requires averaging the kernel with respect to the weights. Here, $\tilde{Q}$ is given by-

$$\tilde{Q} = \sum_{k,k',q,q'} \left(w_k w_{k'} \left|k+k',0\right\rangle + v_k v_{k'} \left|0,k+k'\right\rangle + w_k v_{k'} \left|k,k'\right\rangle + w_{k'} v_k \left|k',k\right\rangle\right) \cdot \\ \cdot \left(\left\langle -q,-q'\right| w_q v_{q'} + \left\langle -q',-q\right| w_{q'} v_q + \left\langle -\left(q+q'\right),0\right| w_q w_{q'} + \left\langle 0,-\left(q+q'\right)\right| v_q v_{q'}\right) \tag{35}$$

Since the action in Fourier space depends only on the magnitude of the weights and not on the phase, we deduce that the phases are uniformly distributed on the unit circle and are independent of the magnitudes. There are multiple types of terms that appear in $\tilde{Q}$, and we will calculate each part individually:

1. For terms of the form- $w_k v_{k'} w_q v_{q'}$, we have-$\langle w_k v_{k'} w_q v_{q'}\rangle = \langle |w_k|\,|v_{k'}|\,|w_q|\,|v_{q'}|\rangle \left\langle e^{i\left(\phi_k+\phi_q+\theta_{k'}+\theta_{q'}\right)}\right\rangle$. Since the phases of v, w are independents, then $\left\langle e^{i\left(\phi_k+\phi_q+\theta_{k'}+\theta_{q'}\right)}\right\rangle = \left\langle e^{i(\phi_k+\phi_q)}\right\rangle \left\langle e^{i\left(\theta_{k'}+\theta_{q'}\right)}\right\rangle$. If $q \neq \pm k$ then $\left\langle e^{i(\phi_k+\phi_q)}\right\rangle = \left\langle e^{i\phi_k}\right\rangle \left\langle e^{i\phi_q}\right\rangle$ and since the phases are uniformly distributed over $[0, 2\pi]$ these averages vanish and if $q = k$ then $\left\langle e^{i(\phi_k+\phi_q)}\right\rangle = \left\langle e^{2i\phi_k}\right\rangle$ which vanishes too, so that $\left\langle e^{i(\phi_k+\phi_q)}\right\rangle = \delta_{k,-q}$. Similarly, $\left\langle e^{i\left(\theta_{k'}+\theta_{q'}\right)}\right\rangle = \delta_{k',-q'}$.Thus, we have-$\langle w_k v_{k'} w_q v_{q'}\rangle = \delta_{k',-q'}\delta_{k,-q} \left\langle |w_k|^2 |v_{k'}|^2\right\rangle$
2. For terms such as- $\langle w_k w_{k'} w_q v_{q'}\rangle$, by averaging over the phases of $v_{q'}$ the term vanishes, and similarly terms like $\langle w_k v_{k'} v_q v_{q'}\rangle$ vanish.
3. We are left with terms of the form- $\langle w_k w_{k'} w_q w_{q'}\rangle$, for this not to vanish, we require $\phi_k + \phi_q + \phi_{k'} + \phi_{q'} = 0$ so that either $k = -k'$ and $q = -q'$, $k = -q$ and $k' = -q'$ or $k = -q', k' = -q$. The same is true for $\langle v_k v_{k'} v_q v_{q'}\rangle$.

In conclusion, the average kernel is given by-

$$\begin{aligned} \sigma_a^{-2} Q = & \sum_{k,k'=0}^{P-1} \left\langle |w_k|^2 |v_{k'}|^2\right\rangle \left|k,k'\right\rangle \left\langle k,k'\right| \\ & + \sum_{k,k'=0}^{P-1} \left\langle |w_k|^2 |w_{k'}|^2\right\rangle \left(2\left|k+k',0\right\rangle\left\langle k+k',0\right| + |0,0\rangle\langle 0,0|\right) \\ & + \sum_{k,k'=0}^{P-1} \left\langle |v_k|^2 |v_{k'}|^2\right\rangle \left(2\left|0,k+k'\right\rangle\left\langle 0,k+k'\right| + |0,0\rangle\langle 0,0|\right) \end{aligned} \tag{36}$$

Since the target has no $|0\rangle$ component, it is orthogonal to the vectors $\langle k+k',0|\,, \langle 0,k+k'|$ and $\langle 0,0|$ so that these terms don't contribute when Q acts on the target. We are left with-

$$\begin{aligned} \sigma_a^{-2} Q \boldsymbol{y}^p &= \sum_{t=1}^{P-1} e^{\frac{2\pi i}{P}pt} Q\,|t,t\rangle = \sum_{t=1}^{P-1} e^{\frac{2\pi i}{P}pt} \sum_{k,k'=0}^{P-1} \left\langle |w_k|^2 |v_{k'}|^2\right\rangle \left|k,k'\right\rangle \delta_{t,k}\delta_{t,k'} \\ &= \sum_{k=1}^{P-1} e^{\frac{2\pi i}{P}pk} \left\langle |w_k|^2 |v_k|^2\right\rangle |k,k\rangle = \sigma_a^{-2}\lambda \boldsymbol{y}^P \end{aligned} \tag{37}$$

Since for all $k \neq 0$ the actions are identical, the mean

$$\lambda = \sigma_a^2 \left\langle |w_k|^2 |v_k|^2\right\rangle = \sigma_a^2 \frac{\int \left(|w_k|^2 |v_k|^2\right) e^{-S_k(|w_k|,|v_k|;a)}}{\int e^{-S_k(|w_k|,|v_k|;a)}},$$